

A Silent Measurement-Mapping Defect in Zero-Noise-Extrapolation Pipelines Changes Which Error-Mitigation Method Appears Best

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Abstract—Quantum error mitigation is routinely benchmarked by comparing estimator accuracy across methods. We report a software-correctness failure mode that can invalidate such comparisons without producing any error, warning, or visibly implausible output. When unitary folding rebuilds a transpiled circuit’s terminal measurements, it can discard the layout-induced mapping from physical qubits to classical bits; and when readout-error-mitigation calibration is indexed by virtual rather than physical qubits, the confusion matrix is inverted against the wrong qubits. We give minimal regression tests built on an exact oracle—global folding at odd scale factors implements $UU^\dagger U = U$, so the *ideal* output distribution must be unchanged—and show that under a non-identity layout the defective pipeline returns a different deterministic bitstring. We then quantify the consequences on a complete, previously frozen benchmark of QAOA MaxCut under two IBM calibration snapshots. Running identical seeds and configurations through the defective and corrected code, the two implementations disagree by 0.006–0.713 in expected cut value, and the mitigation method that appears best changes in 80% of 296 cells. Those changes are not cosmetic: following the defective recommendation costs 0.211 in excess true error, against a best attainable error of 0.389. In the frozen study, 18.8% of estimation cells and 75% of optimisation runs were affected, and the headline conclusion did not survive correction; we retract it. We release the corrected implementation, the regression tests, the prospectively fixed protocol, and all raw data. All results are simulator-only; we make no claim of quantum advantage.

Index Terms—quantum error mitigation, zero-noise extrapolation, readout error mitigation, reproducibility, benchmarking, QAOA

I. INTRODUCTION

Error mitigation is the standard way to extract usable expectation values from noisy quantum processors, and comparative benchmarks—which method, at what cost, on which problem—now guide practice [1]–[3]. Those benchmarks are software. Their conclusions depend not only on the mitigation theory but on whether the pipeline that implements it does what its description says.

This paper reports a defect class in that pipeline, demonstrates it, and measures what it does to a benchmark’s conclusions. The defect is not a statistical artefact of an extreme regime; it is a mapping error that is present at every depth,

every noise level and every shot count, and it is silent: the pipeline emits a well-formed circuit and a plausible number.

Concretely, two things can go wrong once a circuit has been transpiled to a device coupling map. First, transpilation chooses a layout π , so classical bit c records physical qubit $q_{\pi(c)}$. Unitary folding that strips the terminal measurements and rebuilds them with a generic “measure all” operation reinstates the mapping $c \leftarrow q_c$, discarding π . Second, readout-error mitigation calibrates a response matrix per qubit; if calibration circuits are indexed by virtual qubit while the measured bits follow π , the inverse is applied to the wrong qubits.

Contributions

- We identify two silent measurement-mapping defects in a budget-matched ZNE/REM pipeline and give **minimal regression tests with an exact oracle** that detect them independently of noise, shots, or problem (Section IV).
- We **quantify the impact on benchmark conclusions**: on identical seeds and configurations, the defective and corrected implementations disagree enough to change which mitigation method looks best in 80% of cells (Section V).
- We report the **corrected budget-matched comparison** of ZNE, REM and their composition on QAOA MaxCut across six distinct instances and two calibration snapshots with contrasting error-channel mix, with paired tests and multiplicity control (Section VI).
- We **retract** the affected conclusion of our own earlier frozen study and release the corrected artifact (Section VII).

We claim none of the following, all of which are established prior work: that folding leaves state-preparation and measurement error unamplified [4]; that odd integer scale factors avoid partial folds [4]; that circuits should be transpiled before amplification [4]; that shot budget governs which mitigation method wins [2]; or that accuracy gains need not improve downstream decisions [5].

II. BACKGROUND AND RELATED WORK

a) *Zero-noise extrapolation*: ZNE estimates the noiseless expectation value by evaluating an observable at amplified noise levels and extrapolating to zero [6], [7]. Digital unitary folding replaces analogue noise scaling by inserting $U^\dagger U$ pairs [8], and is implemented in Mitiq [9]. Majumdar *et al.* [4] set out best practices for digital ZNE, including transpiling before amplification, preferring odd integer scale factors $\{1, 3, 5, \dots\}$ to avoid partial folds, and composing ZNE with readout mitigation because folding does not amplify SPAM error. We adopt all of these recommendations; they are premises here, not results.

b) *Readout-error mitigation*: Measurement error is frequently the dominant error source for shallow circuits [10], and is corrected by inverting a measured response matrix [10], [11]. We use one specific variant—tensored per-qubit inversion with clipping—and our results should not be read as characterising all readout mitigation.

c) *Benchmarking mitigation*: UNITED [2] unifies ZNE, Clifford-data regression and virtual distillation and benchmarks them as a function of total shots on random circuits and QAOA MaxCut under a trapped-ion noise model; readout mitigation is not among its compared arms. Russo *et al.* [3] test platform-independent mitigation across three hardware providers. Fundamental sampling-cost bounds [12], [13] establish that mitigation is never free, which is why cost-equalised comparison is the meaningful one.

d) *Closest related work*: Köster and Mauerer [14] show that Richardson ZNE can produce *artefactual* improvement: once amplification pushes the signal below the noise floor, the extrapolation collapses into a rescaling of a single noisy measurement. Their artefact is a *statistical-regime* phenomenon that occurs in a correct implementation and is diagnosed by a “garbage-folding” control. The defect we report is a *correctness* phenomenon: it occurs at any depth and any signal level, is invisible to a regime-based control, and is fixed in code rather than by choosing scale factors. The two are complementary failure modes of the same pipeline, and we regard [14] as the closest relative of this work.

III. PROBLEM SETTING

a) *Task*: We use QAOA [15] at depth $p = 1$ for MaxCut on six distinct connected six-vertex graphs (Table I), verified pairwise non-isomorphic. For a graph $G = (V, E)$ the objective is $C(z) = \sum_{(i,j) \in E} \frac{1}{2}(1 - z_i z_j)$ and we estimate $\langle C \rangle$. Because $|V| = 6$, both the exact combinatorial optimum $C_{\max}$ and the exact noiseless value of the ansatz are computable, so “error” is unambiguous.

b) *Three quantities, kept distinct*: We separate (i) *estimation error*, the deviation of an estimate from the exact noiseless value *at the same parameters*; (ii) the *optimisation gap* to a best-known ansatz reference; and (iii) the *gap to the exact optimum*. This paper reports only (i). The ansatz reference in Table I is a multistart numerical best-known value over 60 exact-statevector restarts, **not** a proven global optimum of the ansatz, and is labelled as such in the released configuration.

TABLE I

MAXCUT INSTANCES. THE REFERENCE IS A MULTISTART NUMERICAL BEST-KNOWN VALUE, NOT A PROVEN GLOBAL ANSATZ OPTIMUM. “STARTS AT BEST” IS THE FRACTION OF 60 RESTARTS REACHING IT.

Instance	$ E $	$C_{\max}$	best-known $\langle C \rangle$	ratio	starts at best
prism	9	7	5.9392	0.8485	95%
cycle6	6	6	4.5000	0.7500	100%
k33	9	9	6.2321	0.6925	97%
octahedron	12	8	7.3002	0.9125	100%
path6	5	5	3.8679	0.7736	52%
wheel5	10	8	6.3486	0.7936	48%

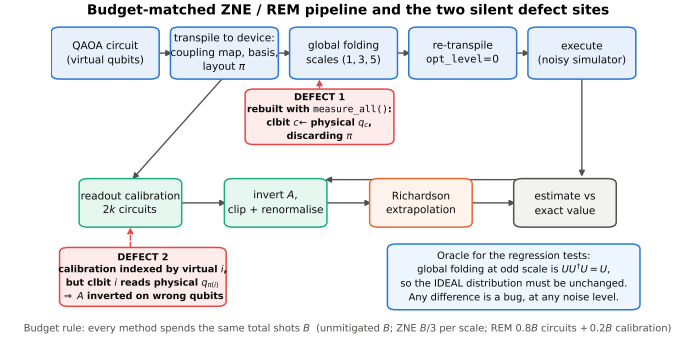


Fig. 1. The budget-matched ZNE/REM pipeline and the two defect sites. Both are silent: the pipeline returns a well-formed circuit and a plausible expectation value.

c) *Noise*: Two static IBM calibration snapshots shipped with `qiskit-ibm-runtime` are used, chosen by measured error rates so that their dominant error channel differs: `dev_lagos` (median readout error 0.169, readout-dominated) and `dev_algiers` (median readout error 0.011, gate-dominated). Both are `cx`-native with at least seven qubits, which holds the folding primitive fixed. An *ideal* (noiseless) condition is included as a reference. All execution is simulated with Qiskit Aer [16]; **no quantum hardware was used**.

d) *Budget matching*: Each estimate is allotted the same total shot budget B regardless of method: the unmitigated arm spends B on one circuit; ZNE spends $B/3$ at each of the scale factors $(1, 3, 5)$; REM spends $0.8B$ on the circuit and $0.2B$ across $2k$ calibration circuits; the composition splits $0.8B$ three ways plus calibration. The realised total is recorded per cell and asserted post hoc; the maximum relative deviation observed across the study was 0.0000%. Calibration policy is held fixed at per-estimate charging inside every comparison, so no comparison is confounded by a change of policy.

IV. THE DEFECT AND ITS DETECTION

A. Defect 1: folding discards the measurement map

Transpiling to a coupling map yields a layout π such that classical bit c records physical qubit $q_{\pi(c)}$. A folding routine that calls a strip-and-rebuild sequence—remove final measurements, fold the unitary, then apply a generic “measure all”—reinstates $c \leftarrow q_c$. When π is not the identity the returned circuit measures the wrong qubits into the wrong bits.

B. Defect 2: calibration indexed by virtual qubits

Readout mitigation estimates a per-qubit response matrix A_q and inverts the tensor product $A = \bigotimes_q A_q$. If the calibration circuits prepare $|0\rangle/|1\rangle$ on *virtual* qubit i while bit i of the measured string comes from physical qubit $q_{\pi(i)}$, then A is inverted against the wrong qubits. On `dev_algiers` with the realised layout, bit 2 was corrected using a response matrix with error rate 0.4638 when the applicable rate was 0.0292—a factor of 15.9—and bit 4 with the reciprocal error.

C. An exact oracle

Both defects are detectable without any noise model. Global folding at an odd scale factor $2k+1$ implements $U(U^\dagger U)^k = U$, so the *ideal* output distribution must be *identical* to that of the unfolded circuit. Any difference is a bug. Using a four-qubit circuit whose ideal outcome is the single bitstring 0101 on a line coupling map:

initial_layout	defective, scale 3	correct
[0, 1, 2, 3]	0101	0101
[3, 2, 1, 0]	1010	0101
[1, 3, 0, 2]	0011	0101
[2, 0, 3, 1]	1100	0101

The identity layout hides the defect entirely, which is why it survives casual testing. Our released test suite additionally checks invariance of exact noiseless $\langle Z_q \rangle$ under folding, operator-level equivalence of the folded core, preservation of classical register structure and of the measurement map, and alignment of calibration circuits with the physical qubits each bit reads.

D. What folding does and does not amplify

For completeness we verify by exact density-matrix computation that gate folding amplifies gate noise ($\langle ZZ \rangle = 0.990000, 0.970299, 0.950990$ at scales 1, 3, 5 under depolarising gate noise) while leaving readout noise *exactly* invariant (1.000000 at all three scales). This confirms rather than discovers the point made in [4]. We also record realised gate counts per scale in every run: at odd scales the error-carrying gates (`cx`, `sx`, `x`) fold a whole number of times, while `rz` — which is error-free in these noise models — does not, so a gate-count ratio alone is necessary but not sufficient evidence of uniform effective noise scaling.

V. IMPACT ON BENCHMARK CONCLUSIONS

We ran identical configurations—same instances, same noise, same budget, same seeds—through the defective and the corrected implementation, and paired the results cell by cell. This isolates the defect from every other source of variation.

Table II reports the paired disagreement. All six tests in the declared family are significant after Benjamini–Hochberg correction ($p_{\text{BH}} \leq 4.5e - 02$), with mean absolute disagreement between 0.006 and 0.713 in expected cut value. The signed mean differences are much smaller than the absolute ones: **the defect does not bias results in a consistent direction, it randomises them**, which is precisely why it is hard to notice.

TABLE II

FAMILY F1: PAIRED DISAGREEMENT BETWEEN THE DEFECTIVE AND CORRECTED IMPLEMENTATIONS ($B = 6000$, SIX INSTANCES $\times$ 30 SEEDS). D IS THE ABSOLUTE DIFFERENCE IN ESTIMATION ERROR. p_{BH} IS BENJAMINI–HOCHBERG ADJUSTED WITHIN THE SIX-TEST FAMILY.

Noise	Method	n	mean $ D $	95% CI	mean sign
dev_algiers	REM	150	0.006	[0.006, 0.007]	-0.00
dev_algiers	ZNE	150	0.463	[0.428, 0.498]	+0.46
dev_algiers	ZNE+REM	146	0.516	[0.481, 0.551]	+0.51
dev_lagos	REM	150	0.442	[0.402, 0.488]	-0.20
dev_lagos	ZNE	150	0.147	[0.133, 0.162]	-0.14
dev_lagos	ZNE+REM	150	0.713	[0.647, 0.783]	-0.43

TABLE III

FRACTION OF CELLS IN WHICH THE BEST-PERFORMING MITIGATION METHOD DIFFERS BETWEEN THE DEFECTIVE AND CORRECTED IMPLEMENTATIONS. *Mean regret* IS THE EXCESS TRUE ERROR (MEASURED UNDER THE CORRECTED IMPLEMENTATION) INCURRED BY FOLLOWING THE DEFECTIVE RECOMMENDATION, AVERAGED OVER FLIPPED CELLS; *best attainable* IS THE MEAN TRUE ERROR OF THE CORRECT CHOICE.

Noise	cells	best changed	rate	mean regret	best attainable
dev_algiers	146	141	96.6%	0.127	0.047
dev_lagos	150	95	63.3%	0.335	0.723
all	296	236	79.7%	0.211	0.389

The practically important consequence is Table III: in 80% of 296 (instance, noise, seed) cells, **the mitigation method that appears best is not the same under the two implementations**. A benchmark run on the defective pipeline would therefore recommend a different method than the same benchmark run correctly, in the majority of cells.

A flip between two near-identical methods would be cheap, so we also report the *regret*: the excess *true* error incurred by following the defective recommendation, measured under the corrected implementation. It is not cheap. On `dev_algiers` the mean regret among flipped cells is 0.127 against a best attainable error of 0.047 — the defective recommendation is several times worse than the correct one — and on `dev_lagos` it is 0.335 against 0.723.

VI. CORRECTED BENCHMARK

Table IV and Fig. 3 report the corrected comparison. Of the 27 tests in family F2, 14 show a practically meaningful improvement over the unmitigated baseline, 2 a practically meaningful degradation, 9 a statistically resolvable effect that falls below the declared practical threshold, and 2 no resolvable difference. We report effects as absolute differences with confidence intervals alongside the problem scale; we do not report ratios alone, because against a small baseline a large ratio can correspond to a negligible absolute effect.

A. Cost accounting

Table V separates circuit shots from calibration shots and circuit executions from calibration executions. Shots are matched across methods; executions deliberately are not, because execution count is the cost the budget rule converts into fewer shots per execution.

Effect of the measurement-mapping defect (simulator only; QAOA $p=1$, 6 instances, budget 6000 shots)

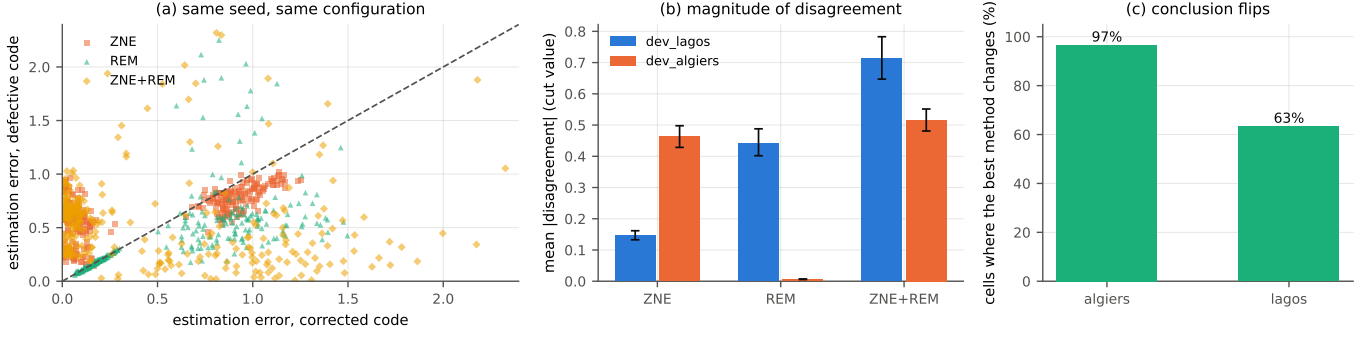


Fig. 2. Effect of the defect, paired on identical seeds and configurations. (a) Per-cell estimation error under the defective versus corrected implementation; points off the diagonal are cells where the defect changed the answer. (b) Mean absolute disagreement with BCa bootstrap 95% confidence intervals. (c) Fraction of cells in which the best-performing mitigation method changes. Simulator only.

Corrected budget-matched comparison (simulator only; shaded = 95% CI of the mean over 6 instances $\times$ 30 seeds)

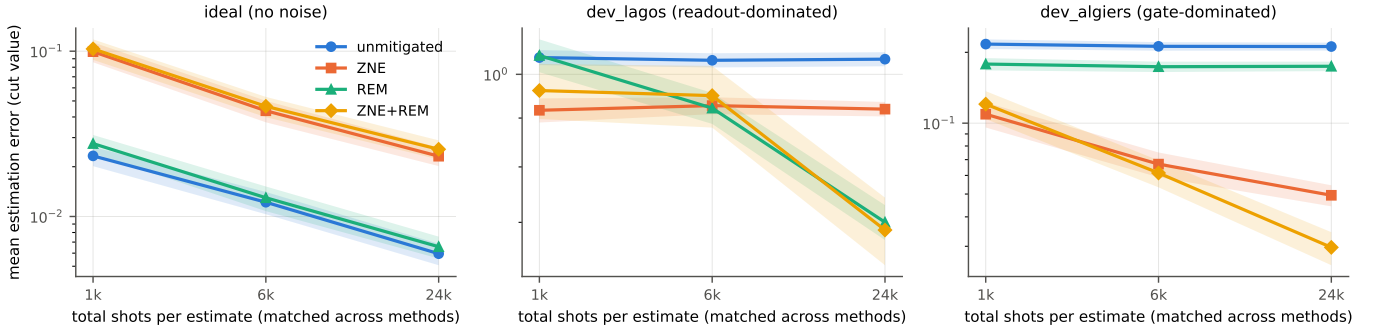


Fig. 3. Corrected budget-matched comparison. Every method spends the same total shots per estimate. Shaded bands are 95% confidence intervals of the mean over six instances $\times$ 30 seeds. Simulator only.

VII. RETRACTION OF AN AFFECTED RESULT

This work began as a correction to our own earlier study, which had been frozen for independent review. Recomputing the realised layout for every condition of that study shows that the defect was active wherever the transpiler chose a non-identity layout: **450 of 2400 estimation cells (18.8%) and 30 of 40 optimisation runs (75%)** used affected code. The unmitigated arm never folds and never calibrates and is unaffected throughout; the ideal and parametric-noise conditions induced no layout and are likewise unaffected.

The earlier study’s headline—that the ranking of mitigation methods reverses between fixed-parameter estimation and in-loop optimisation—rested entirely on runs in the affected set. **We retract it.** We make no in-loop claim in this paper; a corrected in-loop experiment at adequate power is future work.

VIII. THREATS TO VALIDITY

Simulator only. Aer device snapshots omit crosstalk, drift, leakage and non-Markovian effects; nothing here is a hardware result. **Scale.** Six qubits, $p = 1$; mitigation’s exponential

cost [12], [13] is not probed. **Single implementations.** One folding style (global), one extrapolation family (Richardson), one readout-mitigation variant (tensored inversion with clipping); results do not characterise M3, twirled or learning-based readout mitigation, nor local folding. **Device family.** IBM `cx`-native superconducting snapshots only. **Generality of the defect.** We demonstrate it in one pipeline built on widely used components; we have not audited third-party libraries and do not claim they are affected. **Statistical scope.** The declared families F1 and F2 are FDR-controlled; anything else in this paper is exploratory. One arm (the composition) was declared under-powered at the practical threshold in advance.

IX. REPRODUCIBILITY AND DATA AVAILABILITY

All code, the prospectively fixed protocol, configurations, raw per-cell records, processed results and figure sources are in the project repository, together with the regression tests and pinned environment. Every number in this paper is generated from the raw data by the analysis scripts; no result is transcribed by hand. A traceability table maps each

TABLE IV

FAMILY F2: EACH MITIGATED ARM VERSUS THE UNMITIGATED BASELINE AT MATCHED TOTAL SHOTS, PAIRED ON (INSTANCE, SEED). Δ IS THE MEAN PAIRED DIFFERENCE IN ESTIMATION ERROR (NEGATIVE FAVOURS THE MITIGATED ARM); $\Delta/C_{\max}$ EXPRESSES IT IN APPROXIMATION-RATIO UNITS AGAINST THE DECLARED PRACTICAL THRESHOLD OF 0.01. p_{BH} IS ADJUSTED WITHIN THE 27-TEST FAMILY.

Noise	B	Method	$\text{err}_{\text{unmit}}$	$\text{err}_{\text{method}}$	Δ	95% CI	$\Delta/C_{\max}$	p_{BH}	verdict
dev_algiers	1500	REM	0.2167	0.1783	-0.0385	[-0.0444, -0.0335]	-0.0055	3.16e-23	helps (below delta)
dev_algiers	1500	ZNE	0.2167	0.1092	-0.1075	[-0.1220, -0.0933]	-0.0154	2.56e-21	helps (practically meaningful)
dev_algiers	1500	ZNE+REM	0.2167	0.1205	-0.0962	[-0.1125, -0.0785]	-0.0137	1.02e-16	helps (practically meaningful)
dev_algiers	6000	REM	0.2136	0.1753	-0.0383	[-0.0409, -0.0357]	-0.0054	5.69e-25	helps (below delta)
dev_algiers	6000	ZNE	0.2136	0.0675	-0.1461	[-0.1553, -0.1370]	-0.0207	5.69e-25	helps (practically meaningful)
dev_algiers	6000	ZNE+REM	0.2136	0.0614	-0.1521	[-0.1617, -0.1423]	-0.0216	5.69e-25	helps (practically meaningful)
dev_algiers	24000	REM	0.2305	0.1933	-0.0372	[-0.0388, -0.0357]	-0.0050	4.75e-21	helps (below delta)
dev_algiers	24000	ZNE	0.2305	0.0521	-0.1784	[-0.1859, -0.1711]	-0.0238	4.75e-21	helps (practically meaningful)
dev_algiers	24000	ZNE+REM	0.2305	0.0297	-0.2008	[-0.2080, -0.1939]	-0.0268	4.75e-21	helps (practically meaningful)
dev_lagos	1500	REM	1.0413	1.0465	+0.0052	[-0.0317, +0.0473]	+0.0007	9.65e-01	no resolvable difference
dev_lagos	1500	ZNE	1.0413	0.9170	-0.1243	[-0.1481, -0.1008]	-0.0178	5.85e-16	helps (practically meaningful)
dev_lagos	1500	ZNE+REM	1.0413	0.9617	-0.0796	[-0.1388, -0.0170]	-0.0114	3.53e-03	helps (practically meaningful)
dev_lagos	6000	REM	1.0344	0.9218	-0.1126	[-0.1431, -0.0800]	-0.0161	2.94e-10	helps (practically meaningful)
dev_lagos	6000	ZNE	1.0344	0.9274	-0.1070	[-0.1185, -0.0960]	-0.0153	3.90e-24	helps (practically meaningful)
dev_lagos	6000	ZNE+REM	1.0344	0.9500	-0.0844	[-0.1501, -0.0173]	-0.0121	7.69e-03	helps (practically meaningful)
dev_lagos	24000	REM	1.0372	0.7002	-0.3370	[-0.3598, -0.3042]	-0.0481	2.61e-23	helps (practically meaningful)
dev_lagos	24000	ZNE	1.0372	0.9196	-0.1177	[-0.1255, -0.1095]	-0.0168	5.69e-25	helps (practically meaningful)
dev_lagos	24000	ZNE+REM	1.0372	0.6871	-0.3501	[-0.4043, -0.2953]	-0.0500	1.63e-20	helps (practically meaningful)
ideal	1500	REM	0.0233	0.0276	+0.0044	[+0.0020, +0.0065]	+0.0006	1.50e-04	harms (below delta)
ideal	1500	ZNE	0.0233	0.0993	+0.0761	[+0.0635, +0.0903]	+0.0109	3.26e-19	harms (practically meaningful)
ideal	1500	ZNE+REM	0.0233	0.1033	+0.0800	[+0.0667, +0.0953]	+0.0114	1.44e-19	harms (practically meaningful)
ideal	6000	REM	0.0122	0.0130	+0.0008	[-0.0005, +0.0020]	+0.0001	2.63e-01	no resolvable difference
ideal	6000	ZNE	0.0122	0.0436	+0.0314	[+0.0259, +0.0380]	+0.0045	4.89e-20	harms (below delta)
ideal	6000	ZNE+REM	0.0122	0.0464	+0.0342	[+0.0288, +0.0410]	+0.0049	4.75e-21	harms (below delta)
ideal	24000	REM	0.0060	0.0066	+0.0006	[-0.0000, +0.0013]	+0.0001	2.75e-02	harms (below delta)
ideal	24000	ZNE	0.0060	0.0232	+0.0172	[+0.0145, +0.0204]	+0.0025	4.75e-21	harms (below delta)
ideal	24000	ZNE+REM	0.0060	0.0255	+0.0196	[+0.0164, +0.0232]	+0.0028	5.19e-21	harms (below delta)

TABLE V

REALISED COST PER ESTIMATE BY METHOD, CORRECTED IMPLEMENTATION.

Method	total shots	circuit	calibration	circ. exec.	cal. exec.	wall (s)
unmitigated	10,500	10,500	0	1.0	0.0	0.26
ZNE	10,500	10,500	0	3.0	0.0	0.98
REM	10,500	8,400	2,100	1.0	12.0	12.0
ZNE+REM	10,203	8,162	2,041	3.0	12.0	12.0

TABLE VI

TOTAL COST OF THE STUDY. OFFLINE SCORING IS EXACT (STATEVECTOR) AND THEREFORE CONSUMES NO SHOTS; THE EARLIER FROZEN STUDY CHARGED 8,000,000 SHOTS FOR SHOT-BASED SCORING AND DID NOT REPORT THEM.

Quantity	Value
Evaluation cells	6,263
Total shots (circuit)	55,100,400
Total shots (calibration)	6,237,600
Total shots (all)	61,338,000
Offline scoring shots	0 (exact statevector)
Circuit executions	12,789
Calibration executions	39,156
CPU wall-clock	1.37 core-hours

quantitative claim to a raw-data file, configuration, command and output table. The repository is currently private; we will make it public on acceptance, and we do not claim public availability before then.

X. CONCLUSION

Benchmarks of quantum error mitigation are software artifacts, and a mapping defect in the pipeline can change their conclusions without changing their appearance. The defect reported here is silent, present at all noise levels, and detectable by a test that needs no noise model at all: folding at an odd scale factor must leave the ideal distribution unchanged. On a complete benchmark it altered which method appeared best in 80% of cells, and it invalidated the headline of an earlier frozen study, which we retract. We recommend that mitigation pipelines assert the measurement map after any circuit rewrite,

and that readout calibration be indexed by the physical qubits the classical bits actually read.

USE OF AI ASSISTANCE

The implementation, experiments, analysis and drafting of this manuscript were carried out with the assistance of an AI coding agent under author direction. All experimental results were produced by the released code and are reproducible from the released raw data. The authors take full responsibility for the content.

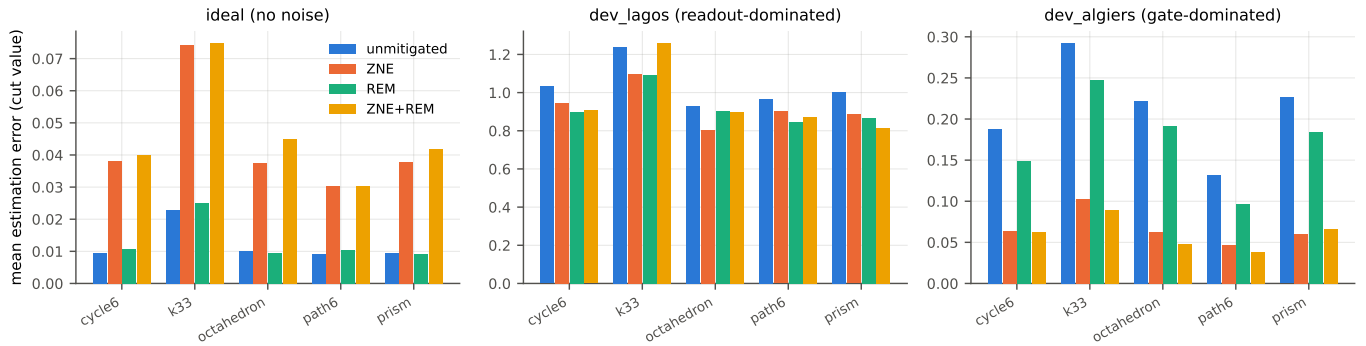


Fig. 4. Per-instance results at $B = 6000$ (30 seeds per bar). Pooled means conceal substantial instance-to-instance heterogeneity. Simulator only.

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